

Chapter 3 Literature Review / Related Work

This chapter reviews the work on which the project builds. It is organised into six categories: deep learning for hip fracture detection and classification; public fracture datasets; the foundations of federated learning; federated learning in medical imaging; personalised federated learning; and detection and segmentation architectures. It then reviews related commercial products and frameworks. Each item is summarised, analysed critically and related to the proposed work, and the chapter ends with summary tables and the research gap.

3.1 Definitions, Acronyms, and Abbreviations

Client drift: The tendency of client models to move towards their local optima during local training, which slows or biases the global model.

DenseNet / Xception: Convolutional neural network architectures widely used as image classifiers.

DRR: Digitally reconstructed radiograph, a synthetic X-ray computed from a CT scan.

Feature shift: Non-IID setting in which clients' images look different (e.g. different scanners) even if labels are distributed alike.

GAN: Generative adversarial network, used here for synthetic data augmentation.

Grad-CAM: Gradient-weighted class activation mapping, a heat-map explanation of a CNN's decision [19].

Label skew: Non-IID setting in which clients have different class distributions or label spaces.

Personalised FL: Federated learning in which each client ends up with a model partly adapted to its own data.

Proximal term: A penalty on the distance between a client's model and the current global model.

3.2 Detailed Literature Review

The review covers 22 research studies and six related applications and frameworks. Studies were selected for their relevance to at least one of three questions: how well can deep learning localise and classify hip fractures on radiographs; how does federated learning behave under heterogeneous clients; and which model families suit per-femur fracture segmentation.

3.2.1 Deep Learning for Hip Fracture Detection and Classification

Early work treated hip fracture detection as image classification; later work added localisation and sub-type classification. The studies below trace this progression and highlight the dependence of all of them

on large single-institution datasets.

3.2.1.1 Detecting Hip Fractures with Radiologist-Level Performance

Summary

Gale et al. [20] trained a DenseNet-based classifier on roughly 53,000 pelvic radiograph studies collected over a decade at one hospital. A first network located the femoral neck region, and a second classified the crop as fractured or not. The system reached an AUC of 0.994, which the authors argued is equivalent to radiologist performance.

Critical analysis

Strengths:

- Very large training set drawn from routine clinical practice.
- Two-stage localise-then-classify design that focuses the classifier on the relevant anatomy.

Weaknesses:

- Binary output only; the subtype, which determines treatment, is not predicted.
- Data from a single institution; the same group later showed that such models can exploit hospital-process confounders (see Section 3.2.1.6).
- Published as a preprint without external validation.

Relationship to the proposed work

This study established that fracture presence can be detected very accurately when data is plentiful, and its localise-then-classify idea is mirrored by our femur ROI detector followed by per-femur analysis. Our work differs by targeting subtypes and by assuming that no single site has tens of thousands of studies.

3.2.1.2 Detection and Visualisation of Hip Fractures on Pelvic Radiographs

Summary

Cheng et al. [21] trained a DenseNet-121 on 3,605 frontal pelvic radiographs after pre-training on 25,505 limb radiographs. On an independent test set of 100 radiographs the model achieved 91% accuracy, 98% sensitivity, 84% specificity and an AUC of 0.98. Grad-CAM heat maps were used for visualisation, and 95.9% of the activation areas lay on the fracture site.

Critical analysis

Strengths:

- Adds a visual explanation, which is important for clinical acceptance.
- Domain-specific pre-training on limb radiographs, useful when target data is limited.

Weaknesses:

- Heat maps are coarse and do not outline the fracture; they are not a segmentation.
- Binary task only, small test set, and a single institution.

Relationship to the proposed work

We share the goal of showing clinicians where the fracture is, but we replace heat maps with explicit polygon masks learnt from radiologist outlines, which can be measured with Dice and IoU. The benefit of domain pre-training supports our plan to share a backbone learnt on all clients before training a small local head.

3.2.1.3 Automatic Hip Fracture Identification and Functional Subclassification

Summary

Kroegue et al. [22] reviewed 1,118 studies and labelled 3,026 hips with bounding boxes in six classes: normal, displaced and non-displaced femoral neck fracture, intertrochanteric fracture, previous internal fixation and previous arthroplasty. An object detector located each hip and a DenseNet classified it. The model reached 93.7% binary accuracy and 90.8% accuracy on the functional subclasses, performed at least at the level of fellowship-trained attending physicians, and improved residents' performance when offered as an aid.

Critical analysis

Strengths:

- Classes chosen for their treatment implications, the same motivation as this project.
- Per-hip analysis, and a reader study showing clinical benefit.

Weaknesses:

- Localisation is a hip box, not a fracture outline; subtrochanteric fractures are not a separate class.
- Single-centre data and a labelling effort that small hospitals cannot repeat.

Relationship to the proposed work

This is the closest single-site analogue of our task. Its per-hip design supports our per-femur format, and

its results give a reference point for subtype accuracy. We extend it by adding the subtrochanteric class and fracture outlines, and by learning across sites whose labels are less complete.

3.2.1.4 Ensemble Classification of Hip Fractures from AP and Lateral Radiographs

Summary

Yamada et al. [23] trained an Xception network on 1,703 AP and 1,220 lateral hip radiographs to classify femoral neck fracture, trochanteric fracture or no fracture. Combining the predictions for both views reached an accuracy of 0.98, comparable to or better than two board-certified orthopaedic surgeons (0.92 and 0.95) and clearly better than two residents.

Critical analysis

Strengths:

- Three-class clinical taxonomy close to ours.
- Shows that a second view adds information.

Weaknesses:

- Requires both views, which our sources do not provide.
- No localisation; single institution; a test set of 150 images per view.

Relationship to the proposed work

The study confirms that the neck/trochanteric distinction is learnable from radiographs. Since our data contains only frontal views, we cannot use its ensemble, and we list lateral views as out of scope.

3.2.1.5 Femoral Neck Fracture Detection and Garden Classification with Advanced Augmentation

Summary

Mutasa et al. [24] used 1,063 AP hip radiographs from 550 patients labelled as Garden I/II, Garden III/IV or normal. A first network localised the femoral neck and a second classified it. Training data was expanded with GAN-generated images and digitally reconstructed radiographs from CT. The system reached 92.3% accuracy for fracture versus no fracture and 86.0% for the three Garden classes; the two-class AUC rose from 0.80 without advanced augmentation to 0.91 with DRRs alone.

Critical analysis

Strengths:

- Directly addresses small datasets through synthetic augmentation.
- Grades fracture severity, not just presence.

Weaknesses:

- Limited to the femoral neck; trochanteric fractures are not covered.
- DRR generation needs CT scans, which most sites do not have alongside radiographs.

Relationship to the proposed work

The large gain from extra data supports our premise that small sites benefit from more (here, federated) data. Our González mapping collapses the Garden I/II and III/IV grades into a single femoral neck class, which is the level of detail the treatment decision needs.

3.2.1.6 Deep Learning Predicts Hip Fracture Using Confounding Variables

Summary

Badgeley et al. [25] showed that deep learning models for hip fracture partly relied on patient and hospital-process variables such as scanner type and priority of the order. When these confounders were balanced in the test set, image-only performance fell to near chance, whereas a model that combined images with the clinical variables explicitly reached an AUC of 0.86.

Critical analysis

Strengths:

- Careful experimental design that exposes shortcut learning.
- Important warning for all single-site medical imaging models.

Weaknesses:

- Does not propose a way to remove confounding from image models.

Relationship to the proposed work

Our clients differ systematically in image source (PACS screenshots, phone photos, Roboflow exports), so a model could learn to recognise the client rather than the fracture. This motivates standardising every client to the same crop size and intensity range, evaluating per client, and testing FedBN, which keeps client-specific normalisation statistics local.

3.2.1.7 Classification and Segmentation of Hip Fractures in X-Rays

Summary

González et al. [5] retrieved 10,308 hip radiographs of 2,618 patients and selected 986 for annotation. Two radiologists outlined every femoral fracture and classified it by region (femoral neck, pertrochanteric, subtrochanteric) and grade. A YOLOv8 segmentation network trained on per-femur crops reached a Dice coefficient of 0.77, 86.2% accuracy for class and grade, and an AUC of 0.981 for fracture detection. The annotated per-femur crops were released as a dataset.

Critical analysis

Strengths:

- First precise segmentation of femoral fractures, which makes the output directly interpretable.
- Public release of radiologist polygons with subtypes and grades.

Weaknesses:

- Single hospital; the subtrochanteric class is very small.
- Neck fractures are harder to outline than trochanteric ones, a pattern we also see in our reproduction (mask AP50 0.298 for neck against 0.824 for intertrochanteric).

Relationship to the proposed work

This study underpins the project. Its per-femur format (1024×1536 crops, YOLO-seg polygons, no label file for no fracture) is the format into which we convert every client, its data is our largest client, and a YOLOv8-seg model trained on it is the helper that proposes missing labels for the other clients. Our reproduction reaches an AUC of 0.974 against the reported 0.981.

3.2.2 Public Fracture Datasets

Public datasets determine what can be studied without access to hospital records. The following dataset provides one of our clients.

3.2.2.1 FracAtlas: A Dataset of Musculoskeletal Radiographs

Summary

Abdeen et al. [12] released FracAtlas, a set of about 4,000 musculoskeletal radiographs from several hospitals in Bangladesh, with image-level fracture labels, bounding boxes and polygon outlines, and per-image flags for the body part, view and hardware. The dataset supports classification, localisation and segmentation benchmarks.

Critical analysis

Strengths:

- Openly licensed, with outlines as well as boxes, and from a South Asian population.
- Rich metadata that allows filtering by view and hardware.

Weaknesses:

- The “hip” flag means that the hip is visible, not that the hip is fractured: of the 338 hip-flagged images, only 63 are fractured and most of those fractures are in the shaft, knee or elsewhere.
- No hip fracture subtypes.

Relationship to the proposed work

FracAtlas is a client with binary labels. Our audit found the mismatch above, so the pipeline keeps only fractures that lie inside a proximal-femur crop, drops non-frontal views (15) and images with hardware (20), and uses non-fractured images as a source of no-fracture femurs.

3.2.3 Foundations of Federated Learning

These works define the algorithms compared in this project and the theory that explains their behaviour on heterogeneous clients.

3.2.3.1 Communication-Efficient Learning from Decentralised Data (FedAvg)

Summary

McMahan et al. [6] introduced federated learning and the Federated Averaging algorithm. In each round, a subset of clients trains the current model for several local epochs, and the server averages the resulting weights in proportion to each client’s data size. Compared with synchronised stochastic gradient descent, FedAvg reduced the number of communication rounds needed by 10 to 100 times on image and language tasks, and it remained workable on unbalanced and non-IID partitions.

Critical analysis

Strengths:

- Simple, general and communication-efficient; the standard baseline.

Weaknesses:

- Convergence slows and the global model can be biased when clients are strongly non-IID.
- Size-weighting lets large clients dominate.

Relationship to the proposed work

FedAvg is our baseline aggregation rule. Its weighting is the reason we study client-size imbalance

explicitly, comparing size-weighted and uniform averaging.

3.2.3.2 Federated Optimisation in Heterogeneous Networks (FedProx)

Summary

Li et al. [9] generalised FedAvg by adding a proximal term $\frac{\mu}{2} \|w - w^f\|^2$ to each client's objective, which keeps local models close to the global model, and by allowing clients to perform different amounts of local work. They proved convergence under bounded dissimilarity and, in highly heterogeneous settings, improved absolute test accuracy by 22% on average over FedAvg.

Critical analysis

Strengths:

- Directly targets client drift with one extra hyperparameter.

Weaknesses:

- μ must be tuned; a large value slows learning at clients with useful local information.
- Evaluated mainly on small benchmark models, not large vision networks.

Relationship to the proposed work

FedProx is one of the three methods we compare. Its proximal term may stop the large González client from pulling the model away from the small institutional client's distribution.

3.2.3.3 Federated Learning on Non-IID Features via Local Batch Normalisation (FedBN)

Summary

Li et al. [8] addressed feature shift, where clients' inputs differ in appearance, for example because they come from different scanners. FedBN keeps each client's batch-normalisation parameters and statistics local and averages all other parameters. The authors gave a convergence analysis and showed that FedBN outperformed FedAvg and FedProx on benchmarks with feature shift, including a multi-site medical dataset.

Critical analysis

Strengths:

- Very small change to FedAvg that targets exactly the type of heterogeneity found between hospitals.

Weaknesses:

- A new site has no local normalisation statistics until it trains, which complicates deployment to unseen sites.

Relationship to the proposed work

Our clients differ strongly in appearance (Roboflow equalisation, phone photographs, PACS exports), so FedBN is a natural candidate. The convolutional backbones considered for this project use batch normalisation throughout, so FedBN can be applied without architectural change.

3.2.3.4 Tackling Objective Inconsistency in Heterogeneous Federated Optimisation (FedNova)

Summary

Wang et al. [26] showed that when clients perform different numbers of local updates, naive weighted averaging converges to the optimum of a mismatched objective. Their analysis covers FedAvg and FedProx, and they proposed FedNova, which normalises each client's update by its amount of local work before averaging.

Critical analysis

Strengths:

- Clear explanation of a hidden bias in FedAvg with unequal clients.

Weaknesses:

- Mainly relevant when local work differs; with fixed local epochs the effect depends on client size.

Relationship to the proposed work

With a fixed number of local epochs, the González client performs about ten times more updates per round than the institutional one, which is exactly the situation analysed. We treat FedNova as an optional fourth method, used if time allows, as it was in Koser et al. [27].

3.2.3.5 Advances and Open Problems in Federated Learning

Summary

Kairouz et al. [28] surveyed the field, distinguishing cross-device from cross-silo federated learning, cataloguing forms of non-IID data (feature, label and quantity skew), and discussing privacy, robustness and fairness.

Critical analysis

Strengths:

- Authoritative taxonomy and a broad list of open problems.

Weaknesses:

- A survey; it gives no specific guidance for medical detection models.

Relationship to the proposed work

Our setting is cross-silo with all three kinds of skew at once: feature skew (image sources), label skew (different label spaces and class mixes) and quantity skew (126 against 1,253). The survey's vocabulary is used throughout this report.

3.2.4 Federated Learning in Medical Imaging

These studies show that federated learning works for medical imaging across real institutions, and they define the gap that this project fills.

3.2.4.1 Federated Learning in Medicine without Sharing Patient Data

Summary

Sheller et al. [29] compared federated learning with centralised training and with simpler collaborative schemes for brain tumour segmentation, using the multi-institutional BraTS data partitioned by the institution each scan came from. Federated learning reached about 99% of the quality of the model trained on pooled data and outperformed the other collaborative schemes tested.

Critical analysis

Strengths:

- A segmentation task, like ours, with data partitioned by real institution.

Weaknesses:

- All institutions shared one label definition and a curated, preprocessed format.

Relationship to the proposed work

This supports the use of FL for segmentation and our comparison with pooled training as an upper bound. Unlike this study, our clients do not share a label definition, which is why we add a harmonisation stage and a two-stage method.

3.2.4.2 The Future of Digital Health with Federated Learning

Summary

Rieke et al. [7] discussed how federated learning could address the data-access problem in digital health, describing aggregation-server and peer-to-peer topologies, the remaining privacy risks, and the technical, legal and organisational challenges of deployment.

Critical analysis

Strengths:

- Clear view of both the promise and the practical obstacles.

Weaknesses:

- Perspective article without experiments.

Relationship to the proposed work

It frames our simulated study as a first step before any real deployment and highlights data harmonisation as a prerequisite, which is the largest part of our first phase.

3.2.4.3 Secure, Privacy-Preserving and Federated Machine Learning in Medical Imaging

Summary

Kaissis et al. [30] reviewed the threats to patient privacy in medical AI, including re-identification and model inversion, and the defences available: anonymisation, federated learning, differential privacy, secure aggregation and homomorphic encryption.

Critical analysis

Strengths:

- Shows that federated learning alone is not a complete privacy guarantee.

Weaknesses:

- Review only; the cost of the stronger defences for large vision models is not quantified.

Relationship to the proposed work

We apply the basic layers it recommends: de-identification of burned-in text, pseudonymised filenames and data that stays local. Differential privacy and secure aggregation are noted as future work.

3.2.4.4 Federated Learning for Predicting Clinical Outcomes in COVID-19

Summary

Dayan et al. [31] trained the EXAM model on chest radiographs and clinical data from 20 institutes

worldwide to predict the oxygen requirements of COVID-19 patients. The federated model achieved an average AUC above 0.92 at 24 and 72 hours, improved average AUC by 16% over locally trained models, and generalised better to other sites.

Critical analysis

Strengths:

- Large, real, global deployment across 20 institutes.

Weaknesses:

- Required substantial infrastructure and coordination, which few groups can provide.

Relationship to the proposed work

Whether a small site gains from federation is the hypothesis we test for the institutional client. The study also used radiographs, showing that FL can cope with differences in X-ray acquisition across sites.

3.2.4.5 Federated Deep Survival Learning for Hip Fracture Risk Prediction

Summary

Koser et al. [27] presented the first federated survival model for hip fracture risk prediction from pelvic radiographs, trained across four clinical sites of the Study of Osteoporotic Fractures and comparing FedAvg, FedBN, FedProx and FedNova.

Critical analysis

Strengths:

- Directly compares the aggregation methods we plan to use, on hip radiographs.

Weaknesses:

- Assumes a uniform label space at all sites, and every client holds more than a thousand images.
- Predicts future risk rather than diagnosing and localising an existing fracture.

Relationship to the proposed work

This is the most closely related federated study, and the gap between its assumptions and our setting motivates the project: our clients disagree on labels and one client is an order of magnitude smaller than the others.

3.2.5 Personalised Federated Learning

When clients differ, a single global model may suit none of them well. Personalised FL gives each client its own adapted model while still sharing what is common.

3.2.5.1 Federated Learning with Personalisation Layers (FedPer)

Summary

Arivazhagan et al. [32] split the network into base layers, which are trained federatedly, and personalisation layers at the top, which each client keeps. On heterogeneous image classification benchmarks this outperformed FedAvg.

Critical analysis

Strengths:

- Simple split between a shared representation and a local head.

Weaknesses:

- Evaluated on classification benchmarks only; the choice of split point is heuristic.

Relationship to the proposed work

Our two-stage method follows the same principle: the backbone is shared and the subtype head is local. Unlike FedPer, our clients also differ in their label spaces, so the local head is needed for another reason too.

3.2.5.2 Exploiting Shared Representations for Personalised Federated Learning (FedRep)

Summary

Collins et al. [33] proposed FedRep, in which clients jointly learn a shared representation while each learns its own head, alternating many head updates with a few representation updates per round. They proved that it recovers the ground-truth representation in a linear setting and showed gains over FedAvg and FedPer.

Critical analysis

Strengths:

- Theoretical support for the shared-representation, local-head design.

Weaknesses:

- Theory limited to linear models; experiments on small classification tasks.

Relationship to the proposed work

FedRep supports sharing the representation and personalising the head. We adopt the simpler sequential form (federated backbone, then local head), and FedRep-style alternation is a possible extension.

3.2.6 Detection and Segmentation Architectures

The architecture fixes how fractures are localised. The works below are the main families of localisation models considered for this project: single-stage detectors of the YOLO family, used for the helper models of the data pipeline, and two-stage detectors with a mask branch. The final model is chosen experimentally.

3.2.6.1 You Only Look Once: Unified, Real-Time Object Detection

Summary

Redmon et al. [10] framed object detection as a single regression from the image to boxes and class probabilities, processing images in real time (45 frames per second) with competitive accuracy.

Critical analysis

Strengths:

- Fast single-stage design that later versions made accurate and easy to train.

Weaknesses:

- The original version struggled with small objects, which later versions improved.

Relationship to the proposed work

The femur ROI detector and the helper segmenter of the data pipeline belong to the YOLO family, chosen for fast training on limited GPU time. Single-stage detectors are also one of the candidate families for the federated backbone, because every federated round multiplies the training cost.

3.2.6.2 Mask R-CNN

Summary

He et al. [34] extended the two-stage Faster R-CNN detector with a mask branch that predicts a pixel mask for each detected object, and it became the standard baseline for instance segmentation.

Critical analysis

Strengths:

- Accurate masks and a flexible, well-understood design.

Weaknesses:

- Slower and heavier than single-stage models, which increases the cost of every federated round.

Relationship to the proposed work

Mask R-CNN is one of the candidate architectures for the federated model, alongside single-stage detectors. Single-stage models are cheaper per round, while two-stage models may localise subtle neck fractures more accurately; the candidates will be compared on the same clients and the best-performing one adopted.

3.2.6.3 Ultralytics YOLOv8

Summary

YOLOv8 [11] is a family of anchor-free detection and instance-segmentation models, released with a Python library for training, validation and export, available in sizes from nano to extra-large.

Critical analysis

Strengths:

- Mature tooling, pretrained weights and a standard polygon label format.

Weaknesses:

- The training loop is not designed for federated learning; it must be wrapped so that weights can be exchanged and batch-normalisation layers selected.

Relationship to the proposed work

The femur ROI detector uses YOLOv8n and the helper segmenter uses YOLOv8s-seg. The library is one of the candidates for the federated model, which will be selected after comparing several architectures.

3.2.7 Related Applications and Frameworks

Several commercial products already detect fractures, and open-source frameworks support federated experiments. They set the benchmark for usability and show which features are missing.

3.2.7.1 Gleamer BoneView

BoneView is a commercial AI system that marks suspected fractures on radiographs of many body regions. In a multi-reader, multi-case study of 480 examinations and 24 readers, AI assistance raised per-patient sensitivity from 64.8% to 75.2% and slightly shortened reading time [16]. It shows the clinical value of AI-assisted fracture reading, but it gives a generic fracture box rather than a hip subtype, and it is trained centrally.

3.2.7.2 AZmed Rayvolve

Rayvolve is a commercial fracture-detection product cleared by the FDA for adult and later for paediatric radiographs [17]. Like BoneView, it covers many body regions and reports fracture locations as boxes. It does not classify hip fractures by treatment-relevant subtype, and its training data is collected centrally.

3.2.7.3 Imagen OsteoDetect

OsteoDetect was the first AI fracture-detection software permitted for marketing by the FDA, in 2018 through the De Novo pathway, and it marks distal radius (wrist) fractures on radiographs [18]. It established the regulatory path for AI fracture aids, but it is limited to the wrist.

3.2.7.4 NVIDIA FLARE

NVIDIA FLARE is an open-source software development kit for federated learning that supports both simulation and real deployment with provisioning, secure communication and privacy filters [35]. It is strong for production medical deployments but heavy for a simulated research study on one machine.

3.2.7.5 Flower

Flower is an open-source framework for federated learning research that separates the server strategy from client code, includes FedAvg, FedProx and other strategies, and offers a simulation engine that runs many clients on one machine [36]. It suits our simulated setting well and can later move to real deployment with the same client code.

3.3 Literature Review Summary Table

Table 3.1 summarises the research studies in the research-project format, and Table 3.2 summarises the related applications.

Table 3.1: Summary of the reviewed research studies: the author and reference, the method and data, the main reported result, and the main limitation with respect to this project.

Author	Method	Results	Limitations
Gale et al. [20]	Two-stage localise-then-classify DenseNet; about 53,000 pelvic studies, one hospital	AUC 0.994, fracture vs. no fracture	Binary only; single site; preprint
Cheng et al. [21]	DenseNet-121 pre-trained on 25,505 limb radiographs, fine-tuned on 3,605 pelvic radiographs; Grad-CAM	Accuracy 91%, sensitivity 98%, specificity 84%, AUC 0.98	Binary; heat maps are not outlines; 100-image test set
Krogue et al. [22]	Hip detector plus DenseNet; 3,026 hips, six functional classes	Binary accuracy 93.7%, subclass accuracy 90.8%	Box-level; no subtrochanteric class; single site
Yamada et al. [23]	Xception; 1,703 AP and 1,220 lateral radiographs; three classes; two-view ensemble	Accuracy 0.98 with both views, surgeon-level	Needs lateral view; no localisation
Mutasa et al. [24]	Two-stage CNN with GAN and DRR augmentation; 1,063 AP radiographs; Garden classes	92.3% (2-class), 86.0% (3-class Garden)	Neck only; DRRs need CT
Badgeley et al. [25]	Analysis of confounding in hip fracture CNNs; multimodal model	Image-only near chance on balanced test; multimodal AUC 0.86	No method to remove confounding from image models
González et al. [5]	YOLOv8-seg on per-femur crops; 986 annotated radiographs; region and grade	Dice 0.77, accuracy 86.2%, AUC 0.981	Single site; few subtrochanteric cases
Abdedeen et al. [12]	Public musculoskeletal dataset with boxes, polygons and metadata	Baselines for classification, localisation and segmentation	“Hip” flag means hip visible; no subtypes
McMahan et al. [6]	FedAvg: local SGD epochs, size-weighted averaging	10–100× fewer rounds than synchronous SGD	Degrades under strong non-IID; large clients dominate
Li et al. [9]	FedProx: proximal term, variable local work	+22% average absolute accuracy over FedAvg in highly heterogeneous settings	Extra hyperparameter μ ; small models
Li et al. [8]	FedBN: local batch normalisation	Outperforms FedAvg and FedProx under feature shift	Unseen sites lack local statistics
Wang et al. [26]	FedNova: normalised averaging of local updates	Removes objective inconsistency from unequal local work	Benefit depends on how unequal local work is
Kairouz et al. [28]	Survey of federated learning	Taxonomy of settings, skew types and open problems	No task-specific guidance
Sheller et al. [29]	FL for brain tumour segmentation, BraTS partitioned by institution	About 99% of pooled-model quality	Uniform labels and curated data
Rieke et al. [7]	Perspective on FL in digital health	Topologies, risks and challenges of deployment	No experiments
Kaissis et al. [30]	Review of privacy threats and defences in medical AI	FL plus complementary defences recommended	No cost analysis for large models
Dayan et al. [31]	FL across 20 institutes, chest X-ray plus clinical data (EXAM)	Mean AUC >0.92; +16% AUC over local models	Heavy infrastructure

Author	Method	Results	Limitations
Koser et al. [27]	Federated survival model on pelvic radiographs, four sites; FedAvg, FedBN, FedProx, FedNova	First federated hip fracture risk model	Uniform labels; >1,000 images per client; risk, not diagnosis
Arivazhagan et al. [32]	FedPer: shared base layers, local personalisation layers	Better than FedAvg on heterogeneous benchmarks	Classification benchmarks only
Collins et al. [33]	FedRep: shared representation, local heads, alternating updates	Gains over FedAvg and FedPer; linear-case guarantee	Theory linear only
Redmon et al. [10]	Single-stage detector (YOLO)	Real-time detection at 45 fps	Early version weak on small objects
He et al. [34]	Two-stage detector with mask branch	Standard instance-segmentation baseline	Heavier than single-stage models

Table 3.2: Summary of related applications and frameworks: their main features, their relevance to this project, and their limitations.

Application	Features	Relevance	Limitations
Gleamer BoneView [16]	Fracture boxes on radiographs of many regions	Shows AI assistance raises reader sensitivity	No hip subtype; centrally trained
AZmed Rayvolve [17]	FDA-cleared fracture detection, adult and paediatric	Commercial benchmark for usability	Boxes only; no subtype; centrally trained
Imagen OsteoDetect [18]	First FDA-permitted AI fracture aid (wrist)	Regulatory precedent	Wrist only
NVIDIA FLARE [35]	FL SDK for simulation and production	Alternative FL framework	Heavy for a single-machine simulation
Flower [36]	Research FL framework with simulation engine and built-in strategies	Candidate framework for our simulation	Custom strategies (FedBN, personal heads) need code
Ultralytics YOLOv8 [11]	Detection and segmentation training library	Used for the pipeline’s helper models; candidate for the federated model	Not federated out of the box

3.4 Conclusion

The literature shows that deep learning can detect hip fractures with high accuracy and, with enough labelled data, classify them into treatment-relevant subtypes and outline them [22, 5]. Every one of these studies, however, relies on a large dataset from one institution, and one of them shows how such models can learn site-specific shortcuts [25]. Federated learning is established for medical imaging [29, 31], and FedProx, FedBN and personalised heads address client drift, feature shift and differing tasks respectively. The only federated hip-radiograph study [27], though, assumes that all sites share one label space and hold over a thousand images each.

No reviewed work combines subtype-level fracture segmentation with federated learning across sites that differ in label granularity and size by an order of magnitude. Commercial products detect fractures

broadly but do not provide hip subtypes and are trained centrally. This project addresses that gap by harmonising heterogeneous sources into one format, sharing a backbone learnt on the task every site supports, and personalising the subtype head at the site that has verified subtype labels.